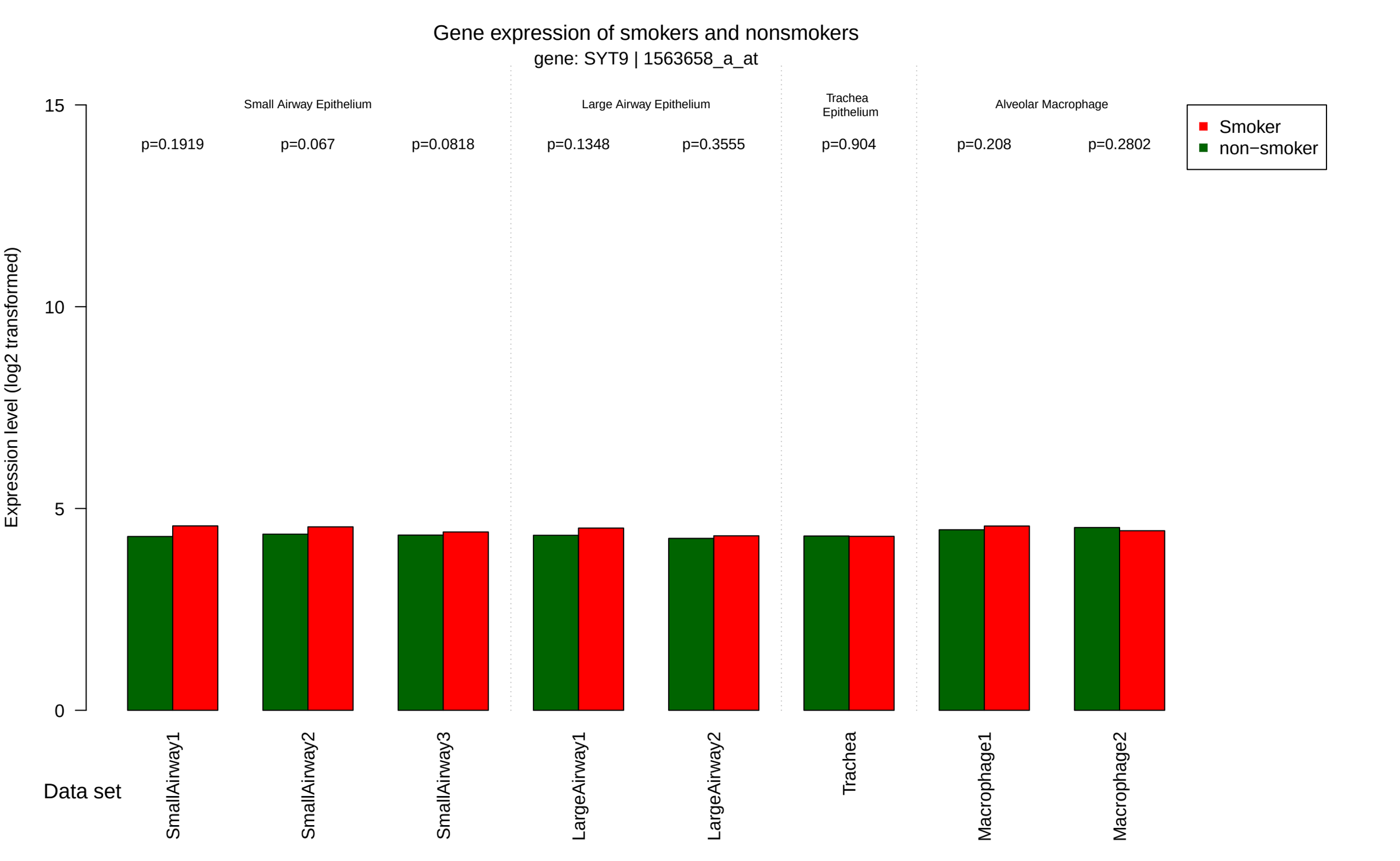
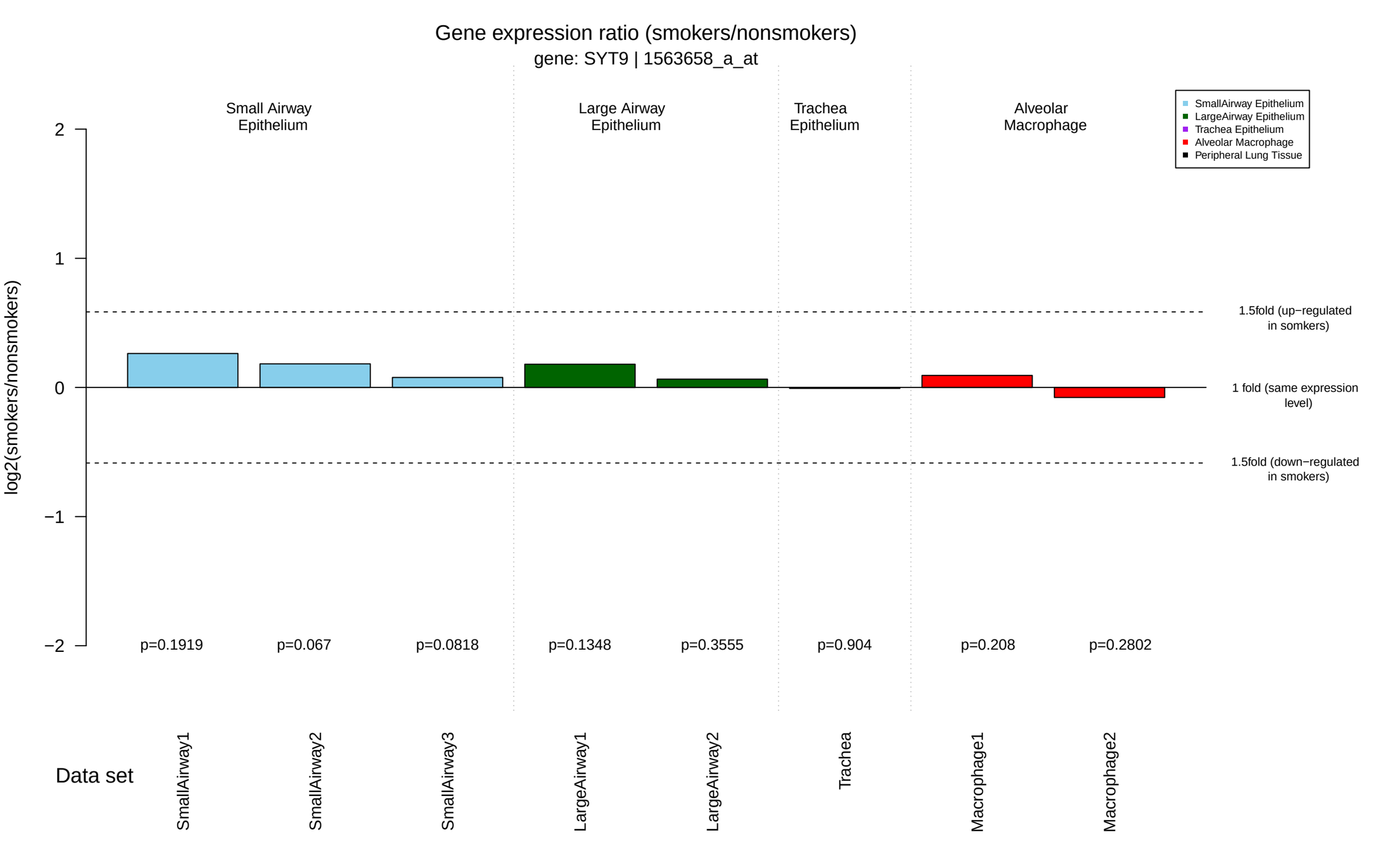
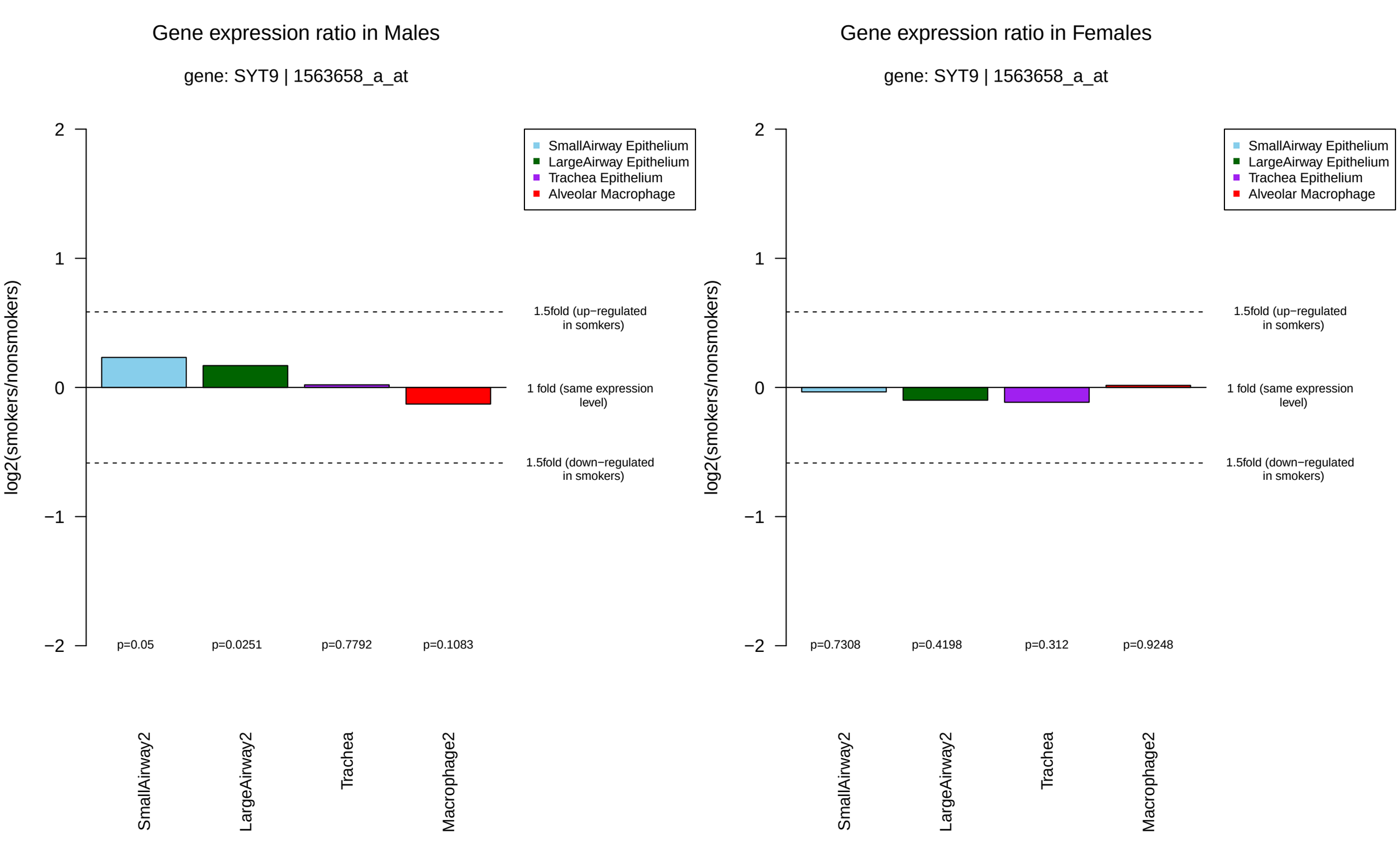
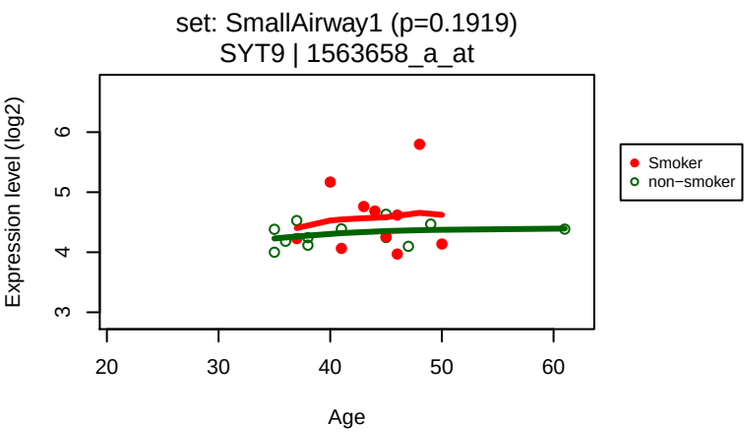
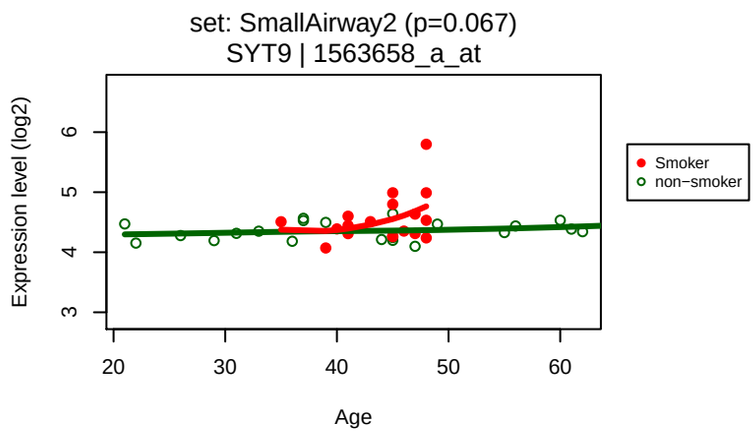
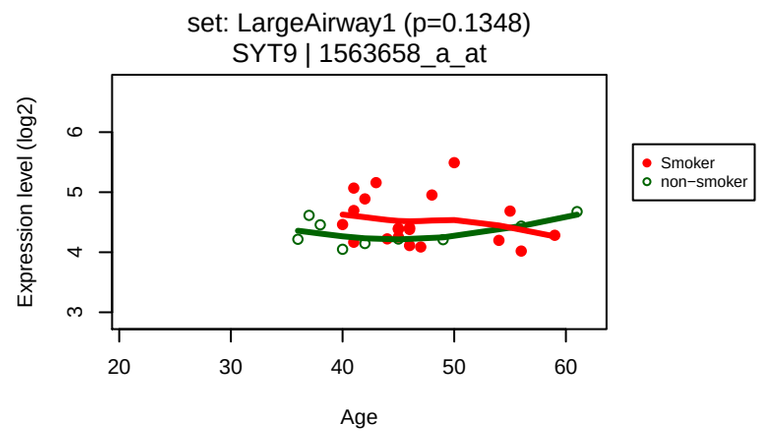

Gene expression of smokers and nonsmokers
gene: SYT9 | 1563658_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1919
p=0.067
p=0.0818
p=0.1348
p=0.3555
p=0.904
p=0.208
p=0.2802
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

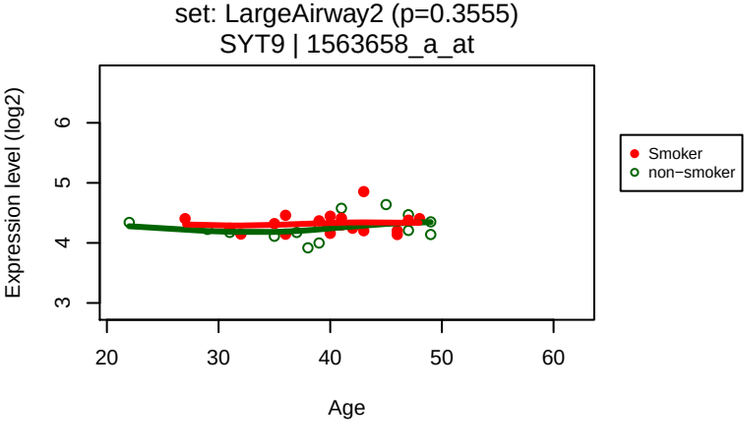

Gene expression ratio (smokers/nonsmokers)
gene: SYT9 | 1563658_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1919
p=0.067
p=0.0818
p=0.1348
p=0.3555
p=0.904
p=0.208
p=0.2802
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

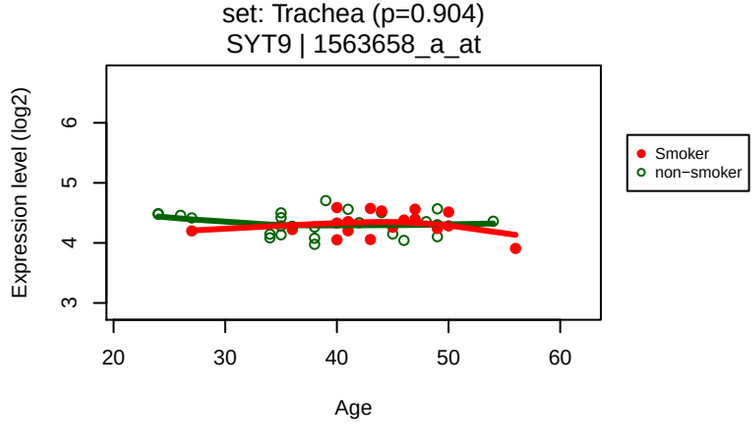

Gene expression ratio in Males
gene: SYT9 | 1563658_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.05
p=0.0251
p=0.7792
p=0.1083
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SYT9 | 1563658_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7308
p=0.4198
p=0.312
p=0.9248
SmallAirway2
LargeAirway2
Trachea
Macrophage2

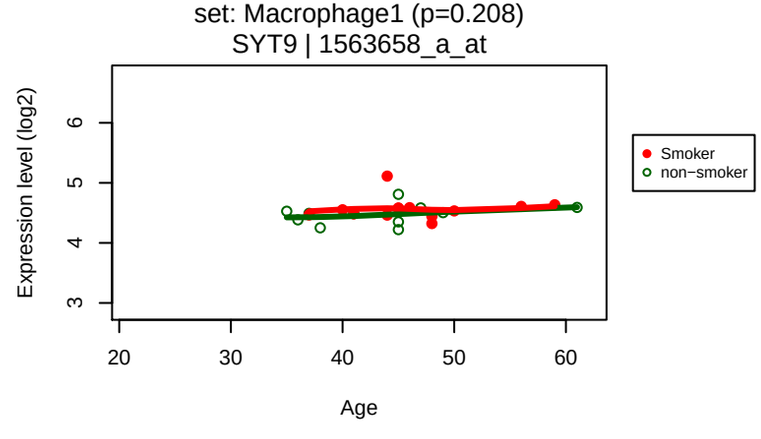

set: SmallAirway1 (p=0.1919)
SYT9 | 1563658_a_at
Expression level (log2)
Age
Smoker
non−smoker

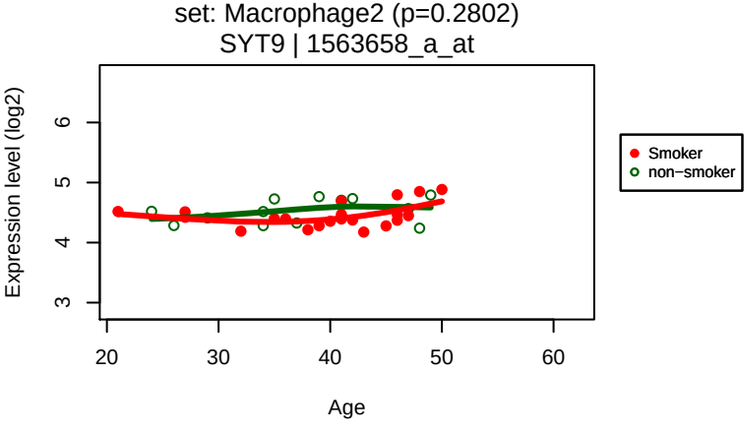

set: SmallAirway2 (p=0.067)
SYT9 | 1563658_a_at
Expression level (log2)
Age
Smoker
non−smoker

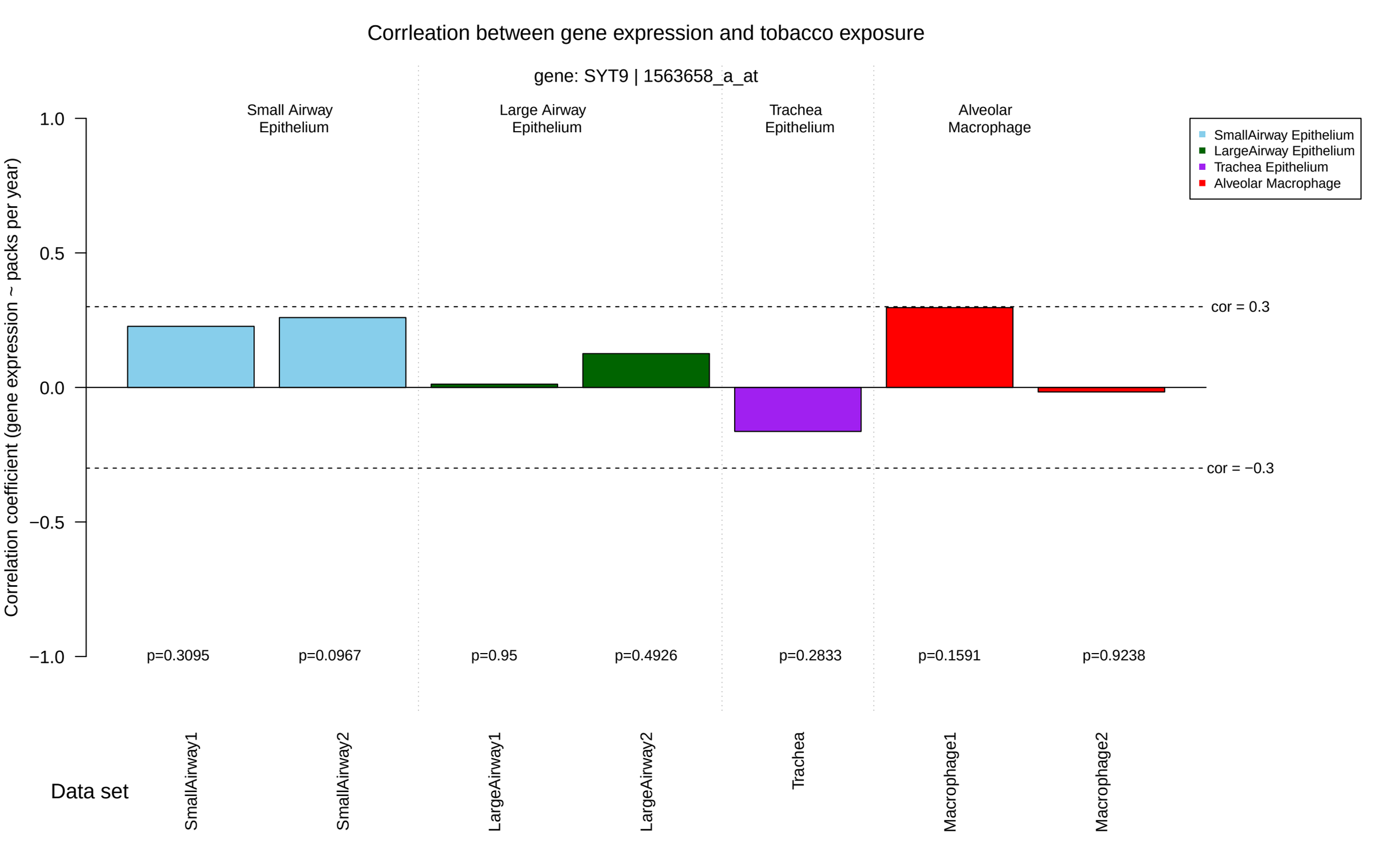

set: LargeAirway1 (p=0.1348)
SYT9 | 1563658_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3555)
SYT9 | 1563658_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.904)
SYT9 | 1563658_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.208)
SYT9 | 1563658_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2802)
SYT9 | 1563658_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SYT9 | 1563658_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3095
p=0.0967
p=0.95
p=0.4926
p=0.2833
p=0.1591
p=0.9238
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2